

DOLITRAIL · CALCULATION NOTE

DoliTrail sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	First issue at TRL 3, on the constructable design of DLT-DDR-002

DoliTrail sizing calculations

On paper, the kit lets two bearers roll a 120 kg patient on the reference doli along a level trail holding about 14 kg each instead of carrying 65 kg each, clamps to poles of 40 to 80 mm without tools, and keeps the doli's own width. The steel frame and clamps carry twice the rated load with a factor of at least 2.0 on yield, and the disc brake has a factor of 1.23 on a wet 30 per cent grade. Four results fall short or are at risk and are posed to Amish as decisions: the wheel and tyre are not catalogue-rated for twice the load (R3), the tyre can slide on wet clay before the brake slips (R5), the kit weighs 14.8 kg in its bag (R7), and two bearers cannot lift the fitted doli over a 400 mm step (R8). The estimated parts cost is USD 455, under the value-engineering target of USD 1,500.

Every figure comes from `docs/04-calcs/sizing.py`, which imports the parametric model (`cad/src/model.py`), so the sizes here are those of the STEP files, the drawings and the build plan. Tags in square brackets match the script output and `docs/04-calcs/results.csv`. These are screening estimates for a paper proof of concept; every one is checked by test at TRL 4.

Safety: DoliTrail carries a person over steep, wet and uneven ground. These numbers are for a design review, not a release for use. No patient is carried until the frame, clamps and wheel have been proof-loaded and the brake has been tested on a wet ramp with a dummy load (DLT-BLD-001, sections 5 and 6).

■ 1. Method and assumptions

The fitted doli is a rigid beam on three supports: the wheel under the patient's hips and the two bearers' pole loops at the ends. Because the bearers choose how much to lift, the wheel takes the weight and the bearers carry only what balances the doli in pitch and roll. Pitch balance comes from the error in placing the wheel under the combined centre of mass; on a slope, from the weight's component along the slope acting at the height of the centre of mass. Brake, grip, clamp friction, structural stress and lift height follow by hand methods in the script.

Table 1. Assumptions.

#	Assumption	Value	Basis
A1	Rated load	120 kg patient on a 10 kg doli; fitted kit 12.3 kg; rolling weight 142 kg	R3; two 60 mm bamboo poles, sticks, cloth and rope
A2	Centre of mass	100 mm above the pole centres; wheel within 100 mm of it	Patient lying on a sagging bed; hip mark on the fitting card
A3	Steadying	Each bearer holds 5 per cent of the rolling weight	Estimate
A4	Tyre grip	0.35 on wet clay or mud (knobbly tyre); 0.60 on a wet timber or concrete ramp	Estimates
A5	Brake	Sintered pads 0.35 wet; 2.0 kN per pad at about 150 N hand pull, held by the parking lock; 203 mm rotor	To confirm with the bought brake
A6	Clamp	Rubber on wet bamboo 0.35; 2.0 kN per clamp (two M8 bolts at 1.0 kN, hand tight)	Estimate
A7	Bamboo	A 40 mm length of 60 mm culm takes 4.0 kN across its diameter in V jaws	To confirm on real poles
A8	Side load	30 per cent of the wheel load at the tyre (side slopes, roots)	Estimate
A9	Steel	ERW tube, yield 210 MPa, E 205 GPa	IS 1161 YSt 210
A10	Wheel and tyre ratings	Wheel 200 kg, tyre 150 kg at maximum pressure	Seller ratings, to confirm
A11	Rolling resistance	0.08 on a dirt trail	Estimate
A12	Masses of bought parts	Wheel 2.60 kg, brake set 0.81 kg, harness 0.75 kg each, bag 0.60 kg, pump kit 0.35 kg	Catalogue values, to confirm

■ 2. Geometry

Table 2. Kit geometry.

Tag	Quantity	Value
[G1]	Wheel diameter, 20 x 2.125 tyre	514 mm
[G2], [G3]	Pole centres and diameters the clamps take	450 to 600 mm; 40 to 80 mm
[G4]	Tyre contact below the pole centres	720 mm
[G5], [G6]	Width on the reference doli; largest pole centres for 700 mm	680 mm; 570 mm
[G7]	Spigot inside its stub at the widest setting	77.5 mm
[Q3]	Cross strap above the tyre	174 mm

■ 3. Mass (R7)

Table 3. Masses.

Tag	Item	kg
[K1]	Fork unit	2.65
[K2]	Side arms, four	3.27
[K3], [K4]	Upper jaws; jaw liners	0.60; 0.21
[K5]	T-handle bolts and set screws	1.05
[K6], [K7]	Wheel; disc brake set	2.60; 0.81
[K8]	Lever mount	0.37
[K9]	Bearer harnesses, two	1.50
[K10]	Straps, bag, card, ties, pump and puncture kit	1.76
[K11]	Kit in the bag	14.83
[K12]	Kit fitted to the doli	12.32

R7 not met: 14.8 kg in one bag. The steel frame with its bolts (7.8 kg) and the wheel and brake (3.4 kg) alone pass 10 kg.

■ 4. Bearer loads (R4)

Table 4. Bearer loads.

Tag	Case	Result
[L1]	Rolling weight	142 kg
[L3]	Carrying today, per bearer: two bearers (four)	65 kg (32.5 kg)
[L4]	Level trail, pitch balance on the heavier end	13.2 kg
[L5]	Level trail, average per bearer with steadying	13.7 kg
[L6]	Reduction against two bearers carrying	79 per cent (R4 met, estimate)
[L7]	30 per cent descent, brake on, downhill bearer	45.5 kg (estimate)
[L8]	Pull to keep rolling on a level trail	56 N each (estimate)
[L9], [L10]	15 per cent side slope: roll moment; force at each hand	170 N m; 154 N (about 16 kg)

■ 5. Brake and grip on a 30 per cent grade (R5)

Table 5. Brake and grip.

Tag	Quantity	Value
[B1], [B2]	Along-slope force; brake torque needed	401 N; 103 N m
[B3], [B4]	Brake torque available, wet; factor	127 N m; 1.23
[B5]	Wheel load normal to the slope	961 N
[B6]	Tyre grip factor on wet clay or mud	0.84
[B7]	Tyre grip factor on a wet test ramp	1.44
[B8]	Shortfall on wet mud	about 7 kg-force

R5 at risk: the brake holds, but on wet clay or mud the tyre would slide before the brake slips; the verification ramp would pass while a real trail might not. Taking part of the weight on the downhill bearer lightens the wheel and makes grip worse, so the shortfall has to be held back along the slope.

■ 6. Clamps (R1)

Table 6. Clamps.

Tag	Quantity	Value
[C1], [C2]	Slip resistance, one clamp; two clamps on one pole	1.98 kN; 3.96 kN
[C3]	Factor on the 1.5 kN pull of R1	2.6
[C4]	Bamboo crushing factor at the clamp force	2.0 (to confirm)
[C5], [C6]	Vertical load per saddle; braking pull per clamp	319 N; 100 N

The V jaws wedge the pole, so the friction force is 2.83 times the bolt force times the friction coefficient. The vertical load is carried in bearing by the saddles, not by friction. **R1 met on paper**, with the crushing capacity of real bamboo to confirm.

■ 7. Structure at twice the rated load (R3)

Table 7. Structure.

Tag	Quantity	Value
[S1], [S2]	Strut axial load; stress with side load	2.51 kN; 104 MPa
[S3], [S4]	Strut factor on yield; buckling factor	2.02; 31
[S5], [S6]	Stub; spigot bending stress	69 MPa; 67 MPa
[S7]	Lowest structural factor on yield	2.02
[S8]	Clamp bolt tension	1.0 kN each (proof load about 8.4 kN)
[S9]	Wheel rating against twice the wheel load	0.70
[S10], [S11]	Tyre rating against twice the wheel load; against the wheel load	0.53 ; 1.05

R3 at risk: the fork unit, arms and clamps meet the 2 to 1 margin; the wheel and tyre do not on their sellers' ratings. Sellers' ratings are working loads with their own unstated margin, so the margin can only be shown by proof-loading the bought wheel.

■ 8. Width, steps and fitting (R6, R8, R2)

Table 8. Width, steps and fitting.

Tag	Quantity	Value
[W1]	Width on the reference doli	680 mm (R6 met)
[H1], [H2]	Rise of the poles for the wheel to clear a 400 mm step; pole height	450 mm; 1,170 mm
[H3], [H4]	Lift per bearer: two bearers; four bearers	71 kg; 36 kg
[T1]	Fitting time, two people	4.7 min (R2 met, estimate)

R8 not met with two bearers: the tyre hangs 720 mm below the poles, so clearing a 400 mm step means lifting the poles from knuckle height to chest height; two bearers would lift about 71 kg each. Fitting time (T1) assumes the wheel travels fitted in the fork unit; the steps are listed in `sizing.py`.

■ 9. Cost (R9)

Value-engineering target: USD 1,500. Estimated cost of the constructable design: USD 455 (USD 1,045 under the target) [Q1], [Q2]. The largest lines are the two harnesses (USD 70), the fork unit (USD 62), the four side arms (USD 60) and the wheel (USD 55).

■ 10. Results against the requirements

Table 9. Results.

ID	Result	Status
R1	Clamp factor 2.6 wet; crushing factor 2.0	Met on paper
R2	About 4.7 min	Met on paper (estimate)
R3	Frame 2.02; wheel 0.70 and tyre 0.53 of twice the load	At risk
R4	79 per cent	Met on paper (estimate)
R5	Brake 1.23; grip 1.44 on a ramp, 0.84 on wet mud	At risk
R6	680 mm	Met on the reference doli
R7	14.8 kg	Not met
R8	71 kg per bearer, two bearers	Not met
R9	USD 455	Under the value-engineering target
R10	One pull on the tab	Met by design
R11	Lever 100 to 150 mm behind the pole loop	Met on paper